

3.3 Subtask 3: Python program - String manipulation (Total: 10 marks; 60 minutes)

In the future, the motor access control gate will be equipped with a feature that allows it to read (Radio Frequency Identification) RFID tags. This means that people who have an RFID tag will be able to gain access through the gate without having to physically stop and pay the fee. To make sure that only authorized individuals are granted access, you will need to create a program in the Python programming language. This program will be used to validate the RFID tag ID (Identification number) that is captured by the gate.

Write a Python program that prompts the user to enter an RFID tag and then validates the tag according to the specified format. If the RFID tag is valid, the program should display the message "RFID tag is valid." If the RFID tag is invalid, the program should display the message "RFID tag is invalid."

Validate an RFID tag with the following criteria:

1. The tag must be exactly 7 characters long.
2. The first character must be a digit.
3. The second character must be a letter.
4. The third to sixth characters must be digits.
5. The last character must be an uppercase "Z".

Review the following examples attentively. Example 1 illustrate the execution of the program:

```
Enter the RFID tag:  
1A2345Z  
RFID tag is valid.
```

```
** Process exited - Return Code: 0 **  
Press Enter to exit terminal
```

Examples 2 – 6 will provide you with a clear explanation for why the RFID tag was deemed invalid.

- A123456Z - The first character is not a digit.
- 1123456Z - The second character is not a letter.
- 12345A6Z - Characters 3 to 6 are not digits.
- 1234567Z - The tag is not exactly 7 characters long.
- 123456z - The last character is not uppercase "Z".

- Save your program, by adding your student number to the file name.
- Compile a word document which includes your student details and the code that you have written.
- Take screenshots of some of the sample runs of your code and include it as part of your document.
- Print your document and submit both the document and the code to your lecturer.